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A comprehensive review on H-type Darrieus wind turbine: aerodynamics, blade profile, CFD simulations

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Abstract

The increasing global need for clean and sustainable energy has led to a surge in research on efficient wind energy conversion systems. The H-type Darrieus wind turbine has drawn a lot of interest among vertical axis wind turbines (VAWTs) because of its straightforward design, acceptance of omnidirectional winds, and adaptability to low-wind and urban settings. The aerodynamic performance of H-type Darrieus wind turbines are thoroughly examined in this review, with particular attention paid to important design elements such blade profile, pitch angle, solidity, number of blades, and tip speed ratio (TSR). The impact of flow separation, vortex formation, and dynamic stall on turbine efficiency is also highlighted in the review. In order to comprehend flow behaviour and optimize performance, both experimental studies and numerical simulations, including those based on computational fluid dynamics (CFD) and blade element momentum (BEM) theory are covered. Additionally examined are recent developments in passive and active flow control methodologies. In addition to identifying future research directions for improving the efficiency and applicability of H-type Darrieus turbines, this study attempts to provide a comprehensive understanding of the aerodynamic factors influencing their performance.

Keywords TSR, Lift Force, Wind Energy, Airfoil, Flow control

Introduction

Wind energy or wind power is utilising wind for affording mechanical power by wind turbines which is later converted into electrical power through generators. It is a renowned, renewable and sustainable sources of energy that possess minimum effect on environment in comparison to burning of fossil fuels [1]. Various advantages are found in wind energy which explores its fastest development globally. It is a clean source of fuel, cost effective, domestic energy source and could be constructed on traditional ranches or farms. Researches have also been undertaken to solve the issues for using wind energy flexibly. Generally, WT (Wind Turbines) are of two kinds as HAWT (Horizontal Axis Wind Turbine) and VAWT (Vertical Axis Wind Turbine). Most of the WTs

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possess horizontal axis that are either downwind or upwind. Though HAWT persist to be renowned for power production in large scale, its equivalent VAWT, particularly Darrieus turbines attained more attention recently on various applications [2]. The Fig. 1 represents a decision-making scenario where an individual is contemplating between two choices: HAWT and VAWT. The HAWT (left side) is described as well-established, commonly used for large-scale power generation. The VAWT (right side) is gaining attention due to its versatility and potential for diverse applications.

Various existing works attempted to enhance the performance of VAWT. Accordingly, a cavity has been created in the blade profile of this turbine. Two dimensional CFD (Computational Fluid Dynamics) have been utilised for simulating turbine and finally compute the forces impacting it. K- ω turbulence model and SST (Shear Stress Transport) has been utilised for completing regulating equations. Impacts of dimple parameters have been considered namely profile, location and diameter for enhancing the aerodynamic performance and efficiency of turbine. Airfoil state has been found to be ideal with a circular dimple having cord length 8% situated on airfoil's pressure side and nearer to leading edge. Finally, efficacy at average coefficients and ideal performance of turbine has been observed to be 25% and 18% when utilising airfoil with cavity in comparison to reference airfoil [2]. Similarly, VAWTs like Darrieus seems to be promising on low wind speed. However, it suffers from minimum efficacy in comparison to HAWT. Comprehensive analysis has been undertaken to improvise the overall performance of this WT. Comparison amongst Gambit meshing and ANSYS Workbench has been undertaken for empirical modelling to conclude overall numerical sequence of this turbine. Subsequently, the sequence of this model has been employed for varied blade airfoils for obtaining better performance. Simulations have been undertaken for different ranges of speed through sliding mesh method relying on URANS turbulent computations. Outcomes reveal that ANSYS Workbench model enhanced by utilising SST model, however it has not been suggested for turbulence models. Besides, CFD method

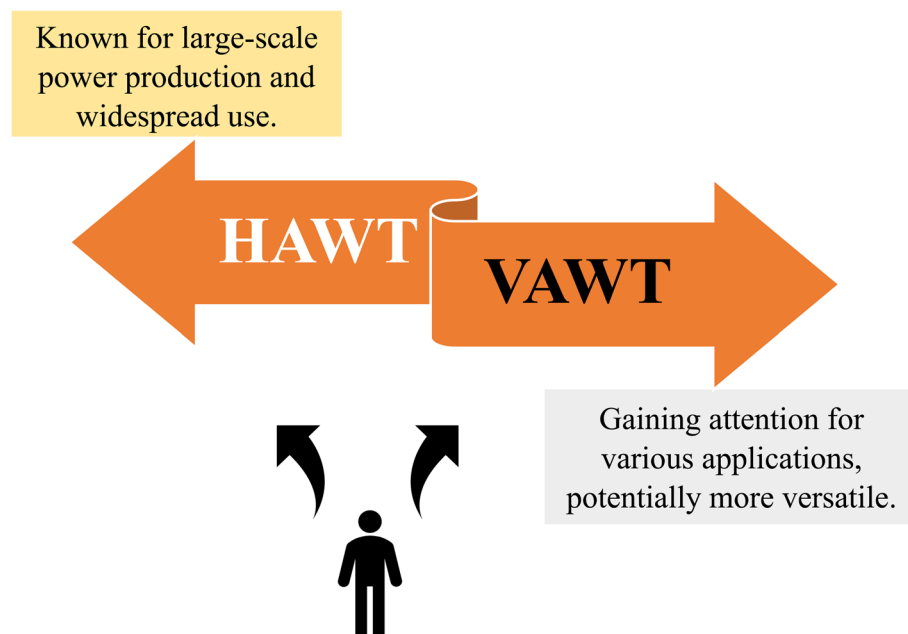


Fig. 1 Decision between horizontal and vertical axis wind turbines

has been utilised in the research for assessing turbine performance having varied airfoil shapes. Results explored high efficacy than typical airfoils at a rate of 10%. Additionally, research has been carried out for analysing this turbine in terms of blade pitch angle. Analytical results revealed that, when pitch angle is zero, best performance is achieved.

All WTs extract wind energy by aerodynamic forces. These forces are of two kinds such as lift and drag. Lift employs force that is perpendicular to relative flow, whereas, drag force employs force on a body in the relative flow direction. It has been stated that GF (Gurney Flaps) could improve the airfoil's aerodynamic performance making them to generate high delay and lift. As its ability was introduced in the initial 70's, it has been employed in various areas including WTs. Probable merits retrieved from GF applications on the DWT blades have been analysed [3, 4]. It indicated an easy way of improving power extraction ability of machines that are experiencing significance within wind energy research [5]. In addition, H-Darrieus, a kind of VAWT has been simulated through 3D URANs for assessing if the numerical model has the capability for adequate prediction of VAWT performance for varied intensities of turbulence. Impact of the turbulence intensity upon power efficiency of H-DWT has been considerable, however, it is limited. Power coefficient enhanced by 22%, whereas turbulence intensity enhanced from 0.7% to 20%. The results verified the impact of dynamic stall through turbulence intensity. However, this impact decreases when flow's Reynolds number over blade enhances [6, 7]. Though traditional works attempted to enhance the performance of H-type DWT, the outcomes varied with respect to the considered parameters. Hence, this study aims to provide a comprehensive review of the H-type DWT by specifically focussing on its aerodynamic performance, blade profile, dynamic stall and other significant parameters to assist researches to develop effective turbine designs in future. The major contributions of this study are shown in Fig. 2.

Genesis of the problem and novelty of the study

Wind energy systems have attracted a lot of interest due to the worldwide focus on sustainable energy. Although horizontal axis wind turbines (HAWTs) dominate the commercial setup, vertical axis wind turbines (VAWTs), specifically that of H-type Darrieus type, have several exclusive benefits, which include ability to accept winds in all directions, low noises, small foot-print design and ease of maintenance. They are

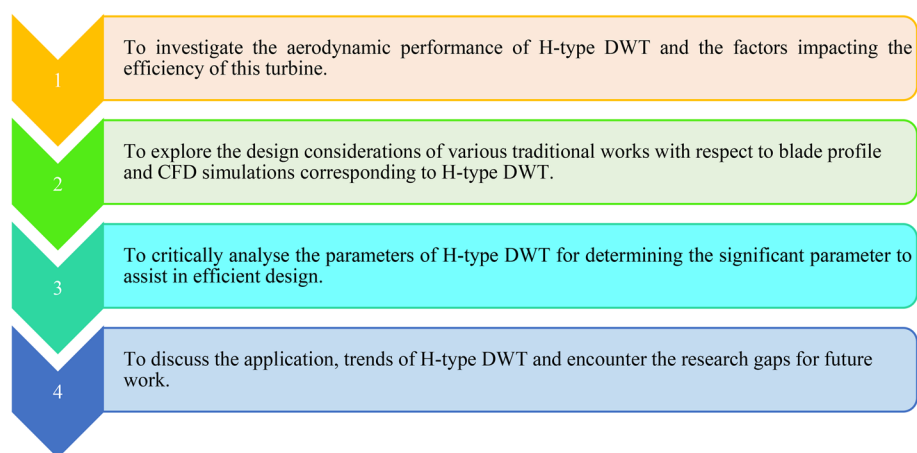


Fig. 2 Objectives of the present studies

nature-friendly, and the features make them fit in the context of decentralized power production in urban and offshore settings. But they have not been successfully deployed in practice because of inherent aerodynamic problems, low self-starting torque, low efficiencies at low tip-speed ratios, and complicated unsteady flow behavior, due to dynamic stalls and dynamic stall interaction. In efforts to counteract such problems, widespread research has been carried out on the optimization of blade geometry, aerodynamic profiling and higher order computational simulation. However, current knowledge is fragmented, with many works looking into specific dimensions of the problem- say blade profiles, rotor configurations or CFD modelling techniques in isolation, but without an understanding of the overall system. This fragmented practice has been an obstacle to the formulation of design guidelines and predictive models that are essential towards better performance of turbines and their expansive use across the world. In a bid to fill these gaps this review provides an integrative and in-depth overview of the H-type Darrieus wind turbine, covering three fundamental areas of investigations; the aerodynamic characteristics of the turbine, blade profile design, and CFD-driven simulation practices. The originality of the research is based on the holistic structure of review where the interconnection between theory and practice has been established. It looks critically at the different airfoil shapes (including NACA series, cambered foils, etc.) adopted in Darrieus turbines, and their effect on the generated torque and efficiency, as well as cutting edge CFD approaches, such as, 2D and 3D modeling, turbulence modeling (RANS, LES), dynamic mesh techniques. Moreover, it presents the areas where the existing practice in the realm of simulation could be enhanced and suggests a research agenda that will help to address these limitations using strategies of experimental validation and more advanced forms of optimization.

Darrieus wind turbine rotors

A Darrieus Wind Turbine Rotor is a type of vertical-axis wind turbine (VAWT) invented by Georges Jean Marie Darrieus in 1931. It is widely recognized for its distinctive egg-beater shaped or H-shaped rotor design as shown in Fig. 3. This turbine comprise of various aerofoil blades fixed on rotating framework or shaft. Blade's curvature permits blades to get stressed only with tension at maximum rotational speeds. Additionally, WTT (Wind Tunnel Tests) finds applicability in assessing the performance of small



Fig. 3 Initial models of H-Darrieus wind rotor

VAWT. Extraction of energy from wind farm or turbine prompts in minimising the speed of upstream-wind. This effect has been named as wind-farm blockage. A technique has been recommended to efficiently correct blockage impact perceived in three WTT undertaken on a DWT. Thus, a single VAWT has been mounted on three tunnels with various test sizes. Angular velocity, torque, power coefficient, TSR (Tip Speed Ratio) and power have been computed. Several traditional and the suggested correction technique have been employed to computed power for compensating blockage effect. Outcomes revealed that the endorsed methodology corrects power coefficient and TSR for DWT in comparison to supplementary correction methods with 30% as blockage ratio [8]. In addition, analysis has been carried out to determine the impact of applying slotted airfoil as blade of turbine on starting and performance characteristics. Specific slot parameters such as inclination angle, dimensions, slot location and parameters have been optimised for the NACA airfoil. This is generally utilised in the Darrieus turbines. ANSYS-FLUENT code has been employed to model flow over turbine. Results explore that VAWT with SA (Slotted Airfoil) possess minimum TSR in comparison to BL (Base-line) turbine which is related with the capability to generate maximum torque at minimum rotational speeds. Analysing aerodynamic performance of SA explores the delay caused by airfoil slot in partition at high attack angles. Thus, it enhances the power coefficient and torque at minimum TSR [9]. Besides, CFD (Computational Fluid Dynamics) has been offering versatile, accurate and cost-efficient means for enhancing the realisation of unsteady aerodynamics corresponding to DWT thereby enhance their efficacy and sound designs. Navier Stokes CFD has been used that features parallel efficiency for comprehensive analysis of DWT blade with respect to 3D unsteady aerodynamics. Clear description and assessment of three major dimensional effects related with periodic-motion of turbine kind has been undertaken involving tip losses, wake/blade interaction and vortex propagation. Results exhibited that 3D flow effects impacting blades of Darrieus rotor have been significantly complex in comparison to the assumption by low fidelity models. It has been frequently utilised for designing that robustly vary during revolution of rotor [10].

Typically, a wind turbine is a non-linear and higher order system which gets easily impacted by disturbances [11]. Wind turbines (WTs) have to operate under stochastic and high vibration wind loads and therefore require sophisticated load control methods. Aerodynamically, performance is highly dominated by the power coefficient (C_p), tip speed ratio (λ), and dynamic stall behavior, and optimum C_p is usually attained in an ideal λ in steady inflow conditions. State Space Control (SSC) and Disturbance Accommodating Control (DAC) are very useful in controlling coupled structural-aerodynamic dynamics and wind disturbance without compromising aerodynamic power capture [12, 13]. Model-based Fuzzy Logic (FL) controllers Two-mass model-based Fuzzy Logic (FL) controllers allow sensorless control and exhibit better transient response and less oscillations than traditional PI controllers [14]. The power optimization, Maximum Power Point Tracking (MPPT) with Tip Speed Ratio (TSR) control, which includes an outer TSR loop with an inner rotor speed loop, operates around the best aerodynamic region and has been shown to be better than the traditional Optimal Torque (OT) approaches [15, 16]. Nonlinearity in aerodynamics associated with increased dynamic stall, wakes interaction and turbulent inflow increases fault vulnerability under extreme conditions [17]. This is alleviated using Fault Tolerant Control (FTC) and Fault Detection

and Isolation (FDI) measures [18, 19]. LPV models with Linear Matrix Inequality (LMI) methods can be applied to detect an error in faults and reconfigure a controller to maintain stability during pitch and torque faults and maintain aerodynamic efficiency [20].

Types of DWT rotors

Darrieus turbine possess minimum initiating torque and is highly efficient. These are VAWTs having low torque and high velocity. It typically comprise of two to three blades. Lift force generated on blade assembly having cross sectional airfoil produces power in DWT. H-rotor turbine is the general kind of Darrieus VAWT which has recently gained more popularity. It is highly suitable when wind speed is low than conventional designs as a result of simple operation, construction and flexibility for facing wind from random direction. A new kind of Darrieus that possess effective starting capability under minimum power coefficient loss has been determined by suggesting a Darrieus turbine having opening. It utilised an airfoil having opening and a typical airfoil situated at trailing end of airfoil. This opening might enhance the torque coefficients during low wind speed. It has also explored that this suggested method might resolve drawbacks of Darrieus turbine (having minimum starting torque) and savonius Darrieus (having more power coefficient loss) [21]. Findings reveal that desirable rotors of inner-opening include rotors with 0.60 and 0.48 as opening ratios. Moreover, rotors having 0.84 and 0.72 as outer ratios seem to be reliable devices and could be employed for replacing the rotor of savonius Darrieus turbine that consider loss of power coefficient, torque coefficient etc. [22, 23]. Various method for describing self-initiation of H-blade rotor has been explored. BEM (Blade Element Momentum) has been compared with 2D and 3D-CFD simulations with respect to TSR, power coefficient and trust force evolution over blade revolution exhibited the strengths and limitations of the suggested method. Start-up ability of H-Darrieus seems to be impacted by several factors like 3D aerodynamic impacts, blade's FAR (Finite Aspect Ratio) and secondary flows. Outcomes revealed the efficiency of endorsed system [24]. Another kind of DWT is J-shaped turbine. Simulation has been undertaken where the J-shaped profile has been utilised as blade airfoil. It has been designed through elimination of pressure side fraction of Du06-W-200 airfoil. Analysis has been undertaken to determine the enhancement of VAWT initiating torque through J-shaped profile. In addition, power-curves for J-shaped and traditional profiles have been computed and torque variation has been attained at various azimuth angles. Pressure field and vorticity around WT has also been explored. Results represented the optimised performance of turbine for the considered profile that removes the airfoil's pressure side from high thickness towards edge of trailing. Besides, through this profile, performance of WT gets intensified and self-initiation of turbine and its TSR has been enhanced [25–27].

Similarly, J-shaped VAWT with rotors of three blade has been analysed. Correct comparisons have been suggested for three varied turbines having various sections of airfoil namely NACAS 1046, NACA 0015 and NACA 0021. This comprised of general airfoil design and J-shape of similar airfoils from two perceptions of WT with respect to noise generation and performance. Complete analysis of performance of suggested design and aero-acoustics have been undertaken through CFD. Two equations namely FW-H (Ffowcs Williams and Hawkins) and URANS (Unsteady Reynolds Averaged Navier Stokes) have been solved concurrently. Impacts of speed ratio, turbine solidity and cut

ratio have been analysed. For all the configurations, outcomes represented that J-shape minimised the performance of wind turbine as it did not minimise noise of turbine. In accordance with the results, J-shape design does not suit well for DWT with respect to reduction of noise generation and turbine performance. Ghafoorian F et al. [28] investigates improving the aerodynamic performance and self-starting capability of hybrid Darrieus-Savonius vertical axis wind turbines (VAWTs) by installing a curtain. CFD simulations were used to determine optimal curtain installation angles, specifically for the upper and lower walls. The findings indicate that the curtain significantly enhances self-starting capability and power coefficient, particularly at low tip speed ratios (TSRs), with optimal upper and lower wall angles of 20° and 45° respectively. This configuration led to a 35% increase in rotor efficiency compared to the solo hybrid rotor.

In addition, a Savonius kind of VAWT has been studied that presents simple construction in comparison to supplementary kinds of VAWTs. A comparative analysis has been carried out for comparing helical turbine to confirm its performance coefficients and aerodynamic characteristics with two stage turbine having similar characteristics. Hence, the main merits of altered kind of helical savonius turbine could be explored and compared with traditional turbine like two stage turbines. Information of trust oriented parameters for this kind of method could afford valuable tool for analysing VAWT functioning in real-time [29]. Likewise, ϕ -shaped DWTs possess high potential claims due to structural merits and Omni-directionality. Yet, the design for these turbines requires an attentive optimisation to accomplish high aerodynamic performance by fitting the wind variations. Optimising shape of this DWT with given inlet-wind speed range has been undertaken. By including HSA (Heuristic Search Algorithm), CMAES (Covariance Matrix Adaptation Evolutionary Strategy) into DMST (Double Multiple Streamtube Model), three geometric rotor variables like blade number, proportion of radius above half of height and equatorial radius are altered in accordance with fitness function which was specifically constructed for satisfying the range requirements of inlet wind. Outcomes explore that this could show an optimised model when range of wind speed is 12.5% enhanced at optimal velocity compared to baseline [30].

H-Type Darrieus rotor aerodynamics

H-rotor system possesses vertical and straight blades rather than standard shapes. It has curved Darrieus turbine blades. Aerodynamic-simulation is difficult for this turbine quantitatively and qualitatively due to dynamic stall and separation of flow throughout the blades. The probable way of enhancing the power output of VAWTs is to analyse its aerodynamic performance without and with winglets. Thus, simulation has been undertaken through ANSYS-CFX software relying on incompressible RANS (Reynolds Averaged Navier Stokes) equation and SM (Structured Meshing) technology. Empirical outcomes and CFD of rotor's power-coefficients have better agreement during range of computations. Power coefficient of this VAWT with winglets have been found to be 10%-19% that is higher than actual H-type VAWT. In individual revolution, output torque corresponding to this turbine has also been larger than actual turbine at different azimuth angle. Torque increases to a maximum of 12.8%, negative torque gets avoided as the TSR (Tip Speed Ratio) attains 1.53. Moreover, the maximum and minimum thrust enhances by 3.4% and 26.7%. In addition, vorticity distribution at tips of the turbine blades have been weaker in comparison to H-rotor having no winglets. It has

been found that aerodynamic performance enhances near the area of turbine blade's tip. Winglets enhance the complete generation of torque, however, it does not enhance at individual azimuth angle for individual blade. Outcomes might afford useful comparison for optimisation and design of further H-rotor VAWTs having winglet [31]. Further, the start-up characteristics of H-type VAWT has been studied by analysing the impacts of physical properties like blade counts and inertia on its power performance through a start-up model. The huge blade counts permits in attaining high power coefficients at TSR less than one that enhances the turbine's self-starting ability [32]. Similarly, a numerical methodology has been suggested to research self-starting characteristics of DWT by considering the moment of inertia of turbine. Simulation initiates from stationary state. It persists until the last steady and periodic condition. In each of time-step, IRV (Instantaneous Rotational Velocity) of turbine has been calculated in accordance with second law of Newton. This has been based on the mechanical and instantaneous aerodynamic forces acting on a turbine. Outcomes represented that, more time has been needed by the turbine to attain its velocity as the inertia of rotor enhances. Turbine might also fail to attain the overall condition which halts the rotation. On contrary, the turbine's oscillation amplitude enhances with decrease in rotor inertia. This could permit turbine for passing high resistant torques in comparison to conventional technique [33, 34]. Besides, H-kind of Darrieus VAWT possess the ability to operate when wind speed is low, however it possess limited aerodynamic performance in comparison to horizontal turbine. Accordingly, effective blade airfoil and wind less configuration has been determined to improve H-rotor's aerodynamic performance. Wind lens has been included in straight bladed turbine. Following this, the impacts of power extension has been analysed through computational techniques. Various configurations of wind lens have been employed to HAWTs. This similarity has also been efficient for H-rotor DWT. It has been found that comprehensive DWT performance could be highly improved through optimisation methods. Eight new airfoil series appropriate for H-rotor DWT have been researched for effective aerodynamic performance. RISø-A1-24, NACA0015, S-1046, FX-L-142 K, DU06-W-200, NACA 63-418, AF300, LS (1)-0417 and FX-L-142 K have been suitable airfoils for individual series of H-rotor DWT. It has been summarised that better performance have been accomplished by symmetric airfoils in comparison to non-symmetric airfoils [35, 36].

Evolution of H-type DWT

Early VAWT have been built based on its aerodynamic drag, while, reverse drag has been removed by encompassing advancing blade for producing net torque. It has also been comprehended that lift force has been efficient parameter than drag. Shape of blade has to be varying on bottom and top to accomplish sufficient lift. Lift based DWT has been introduced by an aeronautical engineer of French named Darrieus in 1925. Though small models were built by Darrieus for establishing his creation, generation model having 7 kW power was constructed by Morel in 1950. Generally, VAWT could be partitioned into savonius rotor, Darrieus rotor, DWT, integrated Darrieus and savonius rotor, sistian kind of wind mill, zephyr turbine and 2-leaf semi-rotary VAWT. In 1931, DWT has been initially designed. Among VAWTs, DWTs explore high efficiency values. As study on VAWT attained momentum during initial 1980s, various configurations were considered and tested. Initially, various failures have been encountered and it was also

restricted to funding in comparison to HAWT. Several innovative ideas have also been not commercialised or unexplored. Over the past thirty years, comprehensive study on the Darrieus rotor resulted in expansion of MST (Megawatt Scale Turbines). Various issues have been solved that was earlier limiting the ability of these turbines during its initial development. Several innovative ideas on the Darrieus rotor strengthen the ability for effective VAWT compared to HAWT. Persisting study on the Darrieus rotor like AHDT has been promising and vital for transformation as device of low wind speed. The limitations of HAWT gets unfold when the size enhances. This leads to transformation for concentrating on DWT rotor as the substitute. On contrary, H-type DWT, a kind of VAWT possess several advantages. It has simplified structure, simple structure and light weight. It also explores high power coefficient of wind and extensive application views [37]. Hence, the VAWT wakes have been researched numerically and empirically with several fidelity degrees like TSR, solidity ratio and Reynolds number. However, numerical and empirical assessment of impacts associated with foil shape in the hydrokinetic turbine wake has been unstudied to find ideal distance amongst two turbines which afford ideal performance. Various simulations have been undertaken based on 2D-3D incompressible VFM (Viscous Flow Model). Effect of foil shape has been analysed with various camber and thickness ratio of similar hydrofoil series of a three bladed H-type DWT. The degree to which it effects the wake structure and hydrodynamic performance of turbine has been assessed. At TSR having blade configuration with high power coefficient, non-symmetrical profile's wake recovery has been effective than symmetrical profile. NACA 4418 and NACA 2418 waves have been similar. But power coefficient corresponding to NACA 4418 has been 4.5% higher when compared to NACA 2418. Thus, outcomes attained from 2D CFD simulations afford valuable and faster data for comprehending their process. Results of this research will also afford few guidelines for enhancing the efficiency of turbine and will support the installation design of turbine farm. NACA 4418 profile has been found to be suitable for application on hydro-farm in a channel flow [38].

Factors impacting the efficacy of H-type DWT aerodynamics

Recently, VAWT has gained high attention as it possesses effective aerodynamic performance when compared to HAWT under complex wind environment. Thus, it possesses various applications in urban environment. In this area, the wind seems to be unsteady and turbulent. Numerical experimentations have been undertaken to analyse aerodynamic performance of self-starting VAWT having various fixed pitch angles and solidities. In this context, incompressible N-S equations integrated with submissive VAWT rotation have been solved. Energy coefficient and flow field have been analysed for varied pitch angle and solidity of turbine. Outcomes revealed that pitch angle and solidity impact VAWT's aerodynamic performance to high extent [39]. In addition, the starting behaviour of H-DWT have been analysed. Basic starting have also been researched. Parameters of rotor geometry have also been researched that regulate the unsteady thrust generation comprising of blade aspect ratio and chord to diameter ratio. Simulation has been undertaken. It has been explored that Darrieus rotor has been susceptible to be sealed in a dead-band having inconsistent thrust throughout blade rotation. This thrust instigated by huge alterations in incidence angle at the start-up lead to inefficiency of Darrieus blade during rotation [40]. Outcomes also exhibit the unsteady thrust

that could be promoted by suitable selection of chord to diameter ratio and blade aspect ratio [41, 42]. Therefore, it has been probable for designing rotors that possess the capability for self-starting. A new self-starting definition has been introduced that rely on its enhanced comprehension. Aerodynamic characteristics of H-type DWT rotor has been analysed. H-rotor turbine having straight blade comprise of two/more blades that have been associated with rotating vertical shaft as presented in Fig. 4. Wind blows above blade's airfoil contours and causes aerodynamic lift thereby pulls blades laterally [43].

Numerical studies have been undertaken on aerodynamic characteristics of VAWT with varied NACA 4 and NACA 6 series foils. Altering the maximum thickness of airfoil, its position, cambers, position and power coefficients under several TSR values have been compared. Power coefficients enhances for symmetrical airfoils having similar and maximum thickness with varied positions and minimises when the position of maximum thickness transfers from leading to the trailing edge [45]. Further, a new design has been suggested for enhancing the performance of turbine and minimise the general drawbacks of these turbines. To accomplish this, a simple porous media configuration has been considered that comprise of a narrow tape. Six varied locations of porous media have been analysed to determine the ideal position of porous media. Subsequently, three bladed Darrieus kind of VAWT having an airfoil profile-DU 06-W-200 in the existence of the porous media have been researched. Outcome have been compared with traditional rotor. Applying porous media assists in the flow for attaching airfoil profile that suppress the partition zone and weaken impact of vorticities. Thus, flow becomes highly stable. Rotor shaft's mechanical loading will also reduce [46]. VAWT has been computationally analysed for improving aerodynamics through leading edge protuberance which could modify from leading edge kind to trailing edge kind [47]. Various other factors affect the aerodynamic performance of H-type turbine. Accordingly, impact of pitch angle on aerodynamics and power performance of VAWT have been studied [48]. Analysis revealed that change in the pitch angle alters the moments and instantaneous loads amongst downwind and upwind splits of turbine. This transformation in the instantaneous moment at revolution for several pitch angles recommend dynamic pitching to be the promising method for optimising the performance further [49]. Impacts of ITV (Instantaneous Tangential Velocity) has been researched on H-type DWT's aerodynamic performance. Outcomes explored the sensitivity of ITV to TSR value and azimuthal

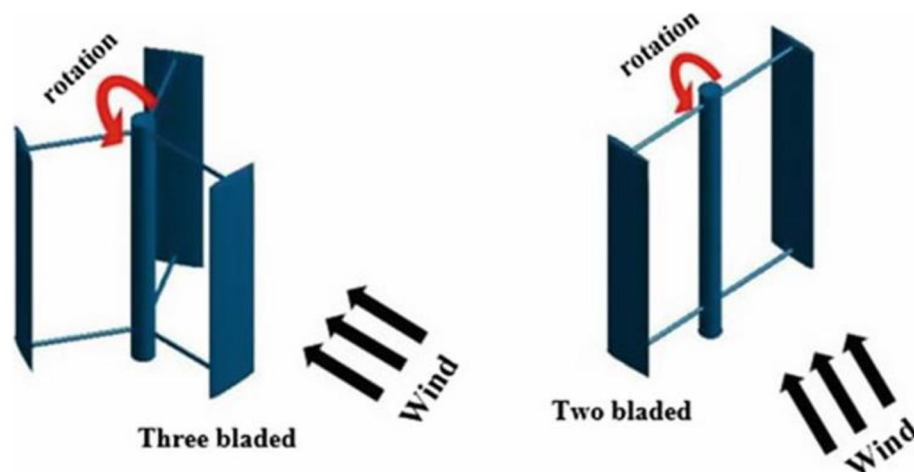


Fig. 4 H-type DWT rotor [44]

position of turbine [50]. Effects of blade thickness on NACA's aerodynamic performance has been studied. Aerodynamic performance degraded at high thickness to chord ratio as huge blade curvature created big and diverging path on blade drag surface [51, 52]. Aerodynamics of low-wind speed corresponding to unsymmetrical H-Darrieus rotor in terms of curvature signature and blade chamber has been compared for enhancing performance [53]. Blade chamber has been suggested to be modest, that is, amongst EN005 and S815 blade for effective design of airfoil [54].

Design considerations of the blade profile

Considering the design parameters of blade contributes substantially in generating improved and innovative VAWT designs. Accordingly, the impacts of the blade attachments on asymmetric H-Darrieus turbine has been analysed when wind speed is low. Power coefficient has been increased by 0.6% possessing attachments on the blade top. Thus, it could be summarised that turbine blade having high thickness degree at top of portion will enhance the turbine's performance [55]. Similarly, performance of VAWT having straight blades have been analysed with respect to power. Field test and WTT (Wind Tunnel Test) has been used for analysis. Test rotor having two blades and NACA0021 profile has been considered. In the field test, wake velocity permits fast recovery compared to wake-velocity in WTT. Moreover, wake velocity reduced with enhancement of TSR [56]. Various traditional works considered different blade profile designs and it is shown in Table 1.

The power coefficient (C_p), self-starting behaviour and optimum tip speed ratio (TSR) of the H-Darrieus wind turbines strongly depends on the selection of the blade profile. As can be seen in the Fig. 5., the comparative studies of the different symmetric, cambered, modified and high-lift airfoils indicate that there is a significant range of response of the turbines as the control conditions change. Widescale two-dimensional CFD screenings of 20–25 airfoil segments have shown that optimised profiles have the potential to increase C_p by about 15–20% over NACA 00xx series blades. S1046 is a common high-performing candidate among symmetric airfoils as a straight blade H-rotor with C_p , max values of about 0.35 or slightly higher in open rotor designs. Cambered airfoils (including S815, EN0005 and NREL S823) tend to be better in self-starting and low-TSR torque, but can have high drag penalties at higher TSRs. An example is that NREL S823 was reported to offer about 6 per cent of higher C_p , max and considerably higher static torque than NACA 0018 in the range of solidities. Blade thickness is also a key aspect with regard to dynamic stall behavior and torque smoothness. The addition of NACA 0012 to NACA 0021 increases the self-starting limit on low wind speeds, but may decrease the maximum C_p on higher Reynolds numbers and TSR values. Thinner airfoils (9–12% thickness-to-chord ratio) tend to give higher peak lift but suffer sudden leading-edge stall, whereas thicker Sects. (18–21%) encourage trailing-edge stall with nicer aerodynamic behaviour, which is more favourable to fatigue life and reduction of torque ripple. Career customizations also improve performance in specific regimes. Blades Slotted NACA 0018 and slotted deflective flaps have been reported to delay flow separation and double low-TSR torque and C_p , with reported increases of 19–27%. Equally, blunt trailing-edge designs are capable of increasing low-TSR C_p by as much as about 33% and have the potential to decrease operational speed and noise, and are thus appealing to urban wind. High-lift profiles including DAE11 and Eppler 423 have shown

Table 1 Design considerations-blade profile

Author	Focus	Technique	Reynolds number	Blade profile	Solidity
Roshan et al. [57]	The study focussed to analyse the performance enhancement of DWT through various cavity layouts	URANS (Unsteady Averaged Navier Stokes) equations have been numerically resolved in integration with SST and K- ω turbulence-model	6.35×10^5	NACA 0021	N/A
Sunny et al. [58]	Existing research focussed on new curved blade VAWTs	WTT	N/A	N/A	N/A
Du et al. [59]	Impacts of several turbine design parameters have been analysed with respect to self-starting behaviour and entire performance	Motor, optical sensor	4.7	NACA4415 DU06W200 NACA0021	0.67 1.0 0.81
Du et al. [60]	Statistic pressure characteristics and time accuracy of blade has been analysed on the rotating H-DWT.	CFD models	N/A	NACA0021	N/A
Hohman et al. [61]	Impacts of inflow-conditions on the VAWT's performance and wake structure has been assessed.	SPIV	2.4 ~ 12.7	NACA0020	0.30
Elkhoury et al. [62]	Performance of O-type (Orthopter-type) VAWT has been analysed.	Hot wire probe, ultrasonic and anemometer	16 ~ 28	NACA0018	1.20 0.67 0.85
V. F. Rolin and F. Porté-Agel [63]	Wake structure has been assessed in the boundary layer	SPIV	1.9	NACA0018	1.08
Bianchini et al. [64]	Analysis has been carried out to determine the optimisation of energy yield of H-DWT in skewed flow	BEM (Blade Element Momentum) method	N/A	NACA0018	N/A
Peng et al. [65]	The study analysed wake and mid-range measurement of the turbine	Hot wire probe Four hole cobra probe	3.4	N/A	1.50
Posa et al. [66]	Wake structure corresponding to isolated VAWT has been analysed	PIV	3.0	NACA0022	1.00
Singh et al. [67]	Overall performance and self-starting behaviour of turbine has been analysed	Anemometer, Dynamometer	2.6 ~ 5.9	S1210	2.40 1.60
Rolland et al. [68]	Model has been validated to analyse the performance of turbine	PIV	N/A	N/A	N/A
Tescione et al. [69]	The wake characteristics have been analysed	SPIV	3.8	NACA0018	0.24
Li et al. [70]	Impact of turbulence intensity, pitch angle, wind speed and blade profile has been analysed to determine the turbine performance	Anemometer, tachometer, induction motor, pitot tube, torque and transducer	0.8 ~ 1.1	NACA0021 NACA634-221 NACA0018	1.50
Greenblatt et al. [71]	Performance of turbine has been analysed with the plasma actuators that have been installed on the leading edges of blade	Plasma actuators and dynamometer	4.4 ~ 7.1	NACA0015	1.25
Gupta and Biswas [72]	Comparative analysis of H-DWT having two and three bladed airfoil has been analysed	CFD analysis	8.5×10^4	NACA0015	N/A

Table 1 (continued)

Author	Focus	Technique	Reynolds number	Blade profile	Solidity
Howell et al. [73]	Impact of blade surface and solidity on the performance of turbine has been assessed	Torque brake and tachometer	2.1 ~ 3.7	NACA0022	1.00 0.67
Gupta and Biswas [74]	Performance computation of H-rotor having three bladed airfoil	CFD and Prandtl lifting line concept	6.06×10^4	NACA airfoil	N/A

*H-DWT (H-Darrieus Wind Turbine)

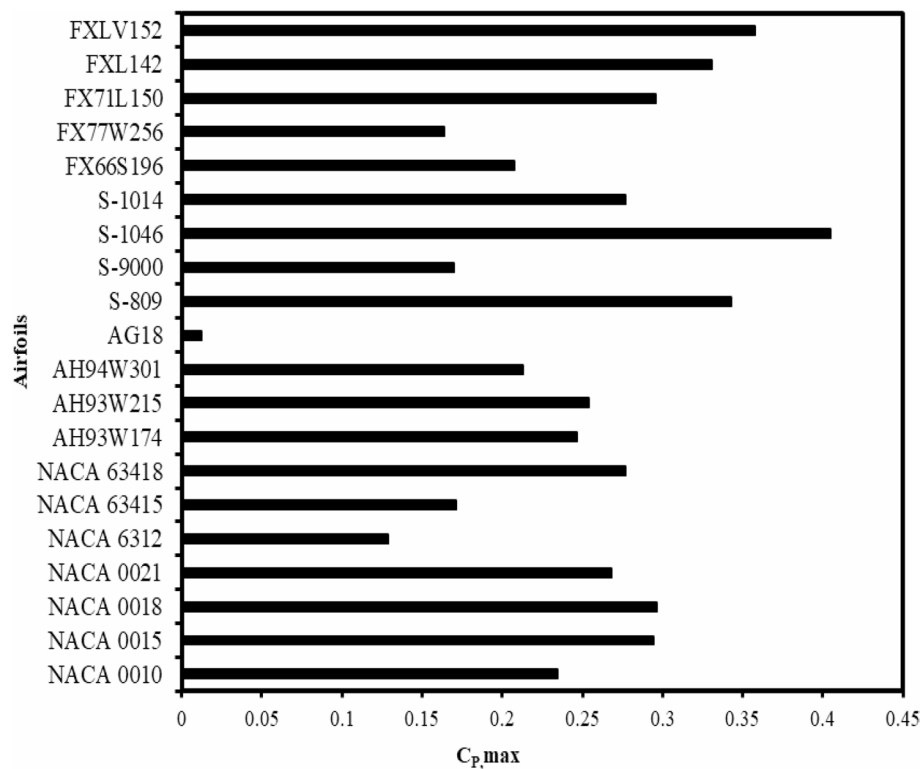
*WTT (Wind Tunnel Test)

*SPIV (Stereoscopic Particle Image Velocimetry)

*PIV (Particle Image Velocimetry)

*CFD (Computational Fluid Dynamics)

*N/A (Not Available)

**Fig. 5** $C_{p,max}$ of various airfoils

$C_{p,max}$ values of over 0.6 in two-dimensional examinations in augmented systems like fairing or wind-lens. The key comparative findings are summarized in Table 2.

CFD model simulations

Simulation using CFD (Computational Fluid Dynamics) has been the significant way of gaining a clear comprehension of the flows of wind farm [75]. Accordingly, an empirical study has been undertaken to assess the aerodynamic behaviour of small-scale VAWT of Darrieus type [76]. Impacts of various design parameters have been assessed namely airfoil type, turbine solidity and number of blades [77]. The K- ϵ model has been selected for performing transient simulations. In addition, MRF (Multiple Reference Frame) model

Table 2 Representative best-in-class blade profiles for H-Darrieus rotors

Objective / Operating Regime	Representative Profiles	Key Performance Outcome
Maximum C_p in open H-rotor	LS(1)0413, S1046, NACA 1418	10–20% C_p improvement over NACA 0018/0021
Strong self-starting, low TSR	S1046, NREL S823, S815, EN0005, NACA 6712	Higher starting torque and improved low-wind performance
Low-TSR enhancement via modification	Slotted NACA 0018, slotted flap, blunt trailing-edge designs	19–33% C_p increase at low TSR
Fairing / wind-lens configurations	DAE11, Eppler 423, S-series high-lift	C_p , max > 0.6 in augmented flow studies

ability of CFD solver has been utilised for expressing dimensionless kind of wind turbine's power output with respect to its rotational speed and stream velocity [78]. Outcomes revealed that optimised turbine explored high power coefficient at a rate of 0.32 and 0.36 in TSR of 3.5 for this simulation and WTT. Analysis has also been undertaken to study H-DWT through traditional Unsteady K-x shear-stress transport-RANS model and CFD [79]. Quantitative influence of 3D impacts associated with end plate usage and blade aspect ratio have been analysed. Outcomes reveal that longer blades enhance the 3D effectiveness. End plates have also been found to possess positive impact on the performance with respect to power extraction based on its limited size and drag [80]. Various CFD simulations undertaken on H-DWT has been summarised and presented in Table 3. However Validation with experimental is a quite important task in the simulation process.

Validation study between numerical results and experiments results of the static torque conducted by [91–94]. Umareddy et al. [93] revealed that several sources of uncertainty can lead to discrepancies between CFD simulations and experimental outcomes, including modelling assumptions, grid independence, boundary conditions, turbulence models, material properties, and scaling effects. Therefore, verification and validation are critical components of CFD research to ensure the reliability and accuracy of the results. In this study, numerical results were validated against wind tunnel experimental data, as illustrated in Fig. 6. The comparison reveals closely matching trends, with only slight deviations in static torque over the rotation cycle.

In the meantime, a standard k- ϵ model has been refined to two prominent alternatives, the realizable k- ϵ model and the renormalization group (RNG) k- ϵ model. In spite of its popularity the standard model fails in flows with high mean shear rates or large-scale separation. It suits best in high Reynold number turbulent flows and does not behave particularly well in low Reynolds number area, like the wall-funtion or transitional regions. In addition to this, its rate of dissipation equation does not contribute to the exactness length scales of turbulence (Figure 7).

In order to eliminate these inadequacies, Shih et al. [95] wrote the realizable k- ϵ model, with modified dissipation-rate formulations, a different expression of eddy viscosity. This is a better model which showed better results on various test cases. To be more physically realistic, the term realizable is used in relation to the fact that the model satisfies mathematical limitation of Reynolds stresses. The realizable k- ϵ model is however not suited to such a suitable zone when there are rotating and impervious fluid boundaries since it is possible in this scenario that a non-physical fluids viscosity may be produced by the model. The RNG k- ϵ model is based on the statistical theory and it has a number of refinements. It takes into consideration the effect of rotation on

Table 3 CFD simulations for analysing H-DWT

S.no	Author	Focus	Turbulence model	Technique	Reynolds number
1.	Patil et al. [81]	Existing research analysed the aerodynamics and flow field of turbine	LES	3D	4.1
2.	Meana-Fernández et al. [82]	Impact of the design parameters on the performance of turbine has been analysed comprehensively	N/A	2D	6.8~10.1
3.	Rezaeiha et al. [83]	Traditional study analysed the effect of operational functionalities on the performance of turbine	SST transition	2.5D and 2D	3.8
4.	Lei et al. [84]	Turbine aerodynamic performance has been analysed in terms of wake features	LES	3D	50.7
5.	Arab et al. [85]	Aerodynamic performance and self-starting features of turbine has been assessed	SST transition	2D	4.7~16.9
6.	Peng et al. [86]	The impacts of turbulence on the aerodynamic performance and wake characteristics of turbine has been examined	LES	3D	3.1
7.	Daroczy et al. [33]	Turbulence models have been comparatively analysed	k-u SST and k- ϵ realisable model	2D	2.0~27.0
8.	Lee et al. [87]	Optimum shape of DWT has been analysed	N/A	3D	8.1~16.2
9.	McNaughton et al. [88]	Various turbulence model has been compared to predict the flow structure of turbine	k- ω SST LRE versions and k- ω SST	2D	2D
10.	Rossetti and Pavesi [24]	Start-up characteristics of turbine has been analysed	SST transition	2D and 3D	3.4
11.	Edwards et al. [89]	Turbulence models have been comparatively analysed with respect to aerodynamic performance	N/A	2D	1.8
12.	Untaroiu et al. [90]	Self-starting characteristics of turbine has been analysed	k- ϵ model	2D and 3D	3.4

* N/A (Not Available)

* LES (Large Eddy Simulation)

* SST (Shear Stress Transport)

eddy viscosity and includes a term on the equation of 8 to assist with fast strained flows. In contrast with the use of constant values offered by a standard model, the RNG model provides analytical representations of turbulent Prandtl numbers. These are improved k-epsilon models used in some past researches.

With respect to CFD methodologies for H-Darrieus rotors, modeling strategies range from two-dimensional unsteady Reynolds-averaged Navier–Stokes (URANS) simulations to fully three-dimensional large-eddy simulations (LES) and fluid–structure interaction (FSI) approaches. Two-dimensional URANS models remain the most widely used technique for parametric investigations due to their relatively low computational cost. Typical configurations employ sliding mesh techniques with k- ω SST turbulence modeling, which has consistently shown better agreement with experimental C_p and wake predictions compared to realizable k- ϵ models. Transition SST and hybrid RANS–LES approaches further improve the prediction of dynamic stall and low-Reynolds-number effects, particularly in micro-rotor applications [10, 80, 96].

The distribution of turbulence and numerical models that are used in simulations of H-Darrieus rotor, as shown in the Fig. 8, clearly shows that they are dominated by RANS-based methods. The k- ω SST model has the largest share (28.5%), as it has a

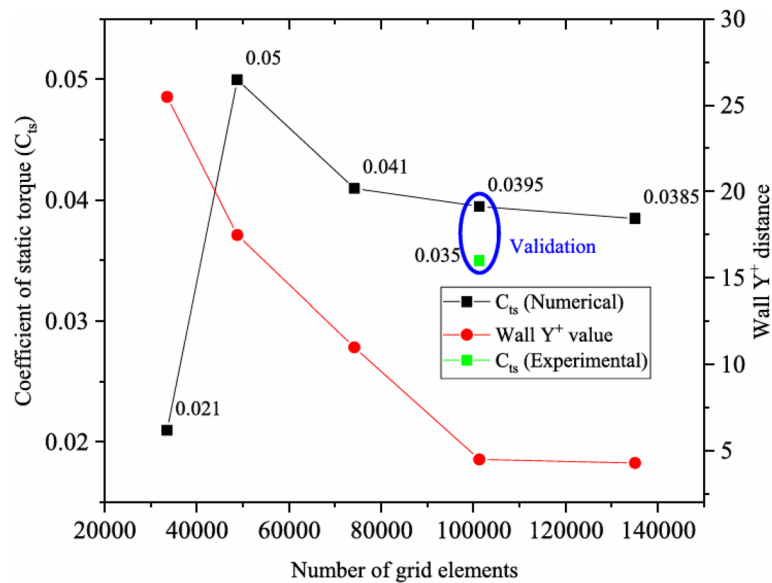


Fig. 6 Validation study of Umareddy et al. [93] (Reproduced with permission)

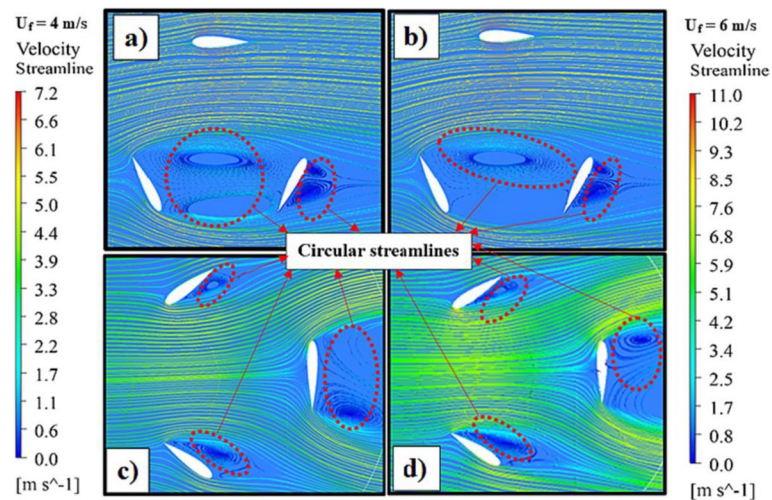


Fig. 7 Streamlines around of NACA 0018 bladed H-rotor. (a) $\theta = 0$, at 4 m/s (b) $\theta = 0$, at 6 m/s (c) $\theta = 30$, at 4 m/s (d) $\theta = 30$, at 6 m/s. (Reproduced with permission)

high ability in addressing adverse pressure gradient and flow separation which are essential in the description of dynamic stall behavior in H-Darrieus rotors. URANS comes close behind (24.1%), which implies that unsteady simulations are important in shedding periodic vortices as well as unsteady torque variations. Standard RANS models (13.9%) and $k-\epsilon$ Realizable model (17.7%), are also popular because of their computational efficiency which is especially useful in preliminary performance predictions. High-fidelity approaches like LES (7.6%), DDES (5.1%) are constrained, however, due to their high computational cost, although they provide a better representation of large-scale turbulent structures and wake dynamics. Lattice Boltzmann Method (3.2%) is the least used and most often can be found in exploratory or scholarly research. In general, almost 84% of presented simulations are based on RANS/URANS formulations, indicating a high level of preference towards cost-effective models, and the issue of scale-resolving

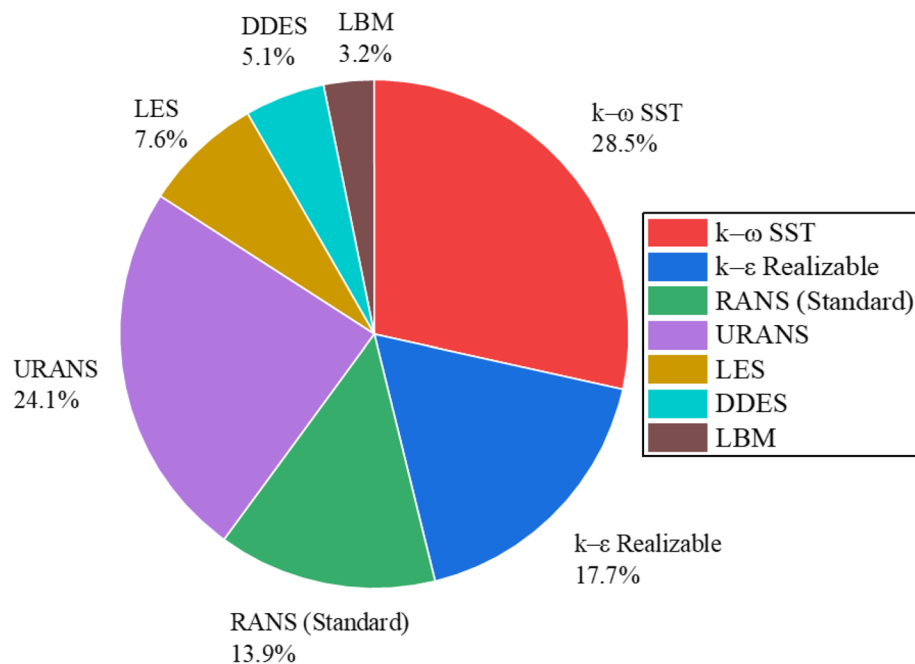


Fig. 8 Distribution of turbulence and numerical models that are used in simulations of H-Darrieus rotor

Table 4 Comparison of 2D and 3D CFD simulation reliability for H-Darrieus rotors

Aspect	2D Simulations	3D Simulations
Typical C_p error vs. experiment	< 15% (well-tuned) to 30–40% (overprediction)	Generally 10–30%, closer to experiments when properly resolved
Physics captured	No tip losses; limited 3D vortex structure	Tip vortices, spanwise flow, full wake development
Computational cost	Low; suitable for parametric sweeps	Very high; requires significant computational resources
Recommended use	Early design, airfoil comparison, trend analysis	Final design validation, load prediction, wake and farm studies

methodology underutilization indicates a large potential scope of more advanced turbulence modeling that should produce more comprehensive foams of vortex interaction and deep dynamic stall behavior unique to H-Darrieus rotors.

The reliability of two-dimensional versus three-dimensional simulations

In CFD, dimensionality significantly affects prediction reliability. Two-dimensional simulations neglect tip losses and spanwise flow, often leading to C_p overprediction and inaccurate wake vortex representation. Comparative 2D–3D studies reveal notable discrepancies in $C_{p, \max}$ and wake topology, particularly at moderate to high TSR values. While well-calibrated 2D URANS models can reproduce general performance trends and are suitable for early-stage design and comparative analysis, three-dimensional simulations are essential for accurate prediction of absolute performance, blade loading, wake interaction, and farm-level assessments [96–98]. The reliability comparison between 2D and 3D simulations is presented in Table 4.

Limitations

In the case of H-Darrieus CFD simulations, model validation is crucial since majority of the RANS commonly used models have significant and well documented limitations, especially when dynamic stall, vortex shedding and wake evolution is to be predicted [80]. Validation is usually done by direct comparison of projected C_p -TSR curves and torque coefficients with wind tunnel or tow tank data. The computational setup is usually tuned to allow errors in prediction of power coefficients to be kept to a level of less than 5–10% when the computational setup is set up carefully and that is with proper domain size, mesh refinements, time-step dependence and parasitic torques corrections. Nevertheless, according to critical reviews, it is inappropriate to associate model reliability only with matching the power coefficient. It should also be accurately validated by comparing the structures of wake vortices, pressure fields, dynamic stall signatures and vorticity fields, particularly in comparing the differences between two-dimensional and three-dimensional models or in choosing between different turbulence closures. Also, some literature confirms complete CFD techniques instead of using isolated simulation cases. This wider validation methodology usually entails grid and time sensitivity analysis in a systematic way, comparison against various experimental data and cross-compensation with more faithful Navier-Stokes models or actuator-line models [99, 100].

Numerical uncertainties

The extensive validation steps are useful in determining numerical uncertainties and biases in the turbulence models, and they boost performance prediction trust. However, pitfalls still exist: excessive fine-tuning of models simulates one experimental C_p vs. TSR curve to fit a particular experimental C_p vs. TSR curve, and when the same model is used with a different geometry or operating point, gives false results. Moreover, uncertainty in numerical results for H-Darrieus CFD arises from discretization, turbulence modeling, boundary conditions, geometric simplifications, and experimental reference data. Reliable simulations require systematic grid and time-step convergence studies, turbulence model comparison, sensitivity analysis, and comprehensive validation against multiple experimental metrics rather than reliance on a single performance parameter [101–104]. Large comparative studies demonstrate that simulation results vary significantly depending on the turbulence model selected [102]. Models such as Spalart–Allmaras (SA), RNG k – ϵ , realizable k – ϵ , and k – ω often fail to accurately reproduce VAWT power curves and blade load distributions across different operating regimes, leading to large errors in C_p and stall onset prediction [93]. Among two-equation models, the fully turbulent k – ω SST formulation is generally considered the most robust; however, it still tends to delay flow separation and under-resolve dynamic stall phenomena, often resulting in C_p overprediction near peak TSR conditions. Fundamentally, RANS models rely on an isotropic eddy-viscosity assumption, which is poorly suited for the strongly separated, rotating, and highly anisotropic flow structures present in H-Darrieus wakes and during deep stall events [105, 106]. There is transition and hybrid RANS in the case of transition and hybrid RANS. The boundaries of laminar separation and stall processes are more accurately represented by Transition SST-based formulations and lead to lower C_p overprediction, especially at low Reynolds numbers. But two-dimensional URANS is not able to model inherently three-dimensional observations like tip vortices and span-wise flow. Hybrid DES/DDES methods enhance the resolution of unsteady vortices at a

very high cost of computation and refined the near-wall grids, which is not very practical [105, 106].

Critical and SWOT analysis

An analysis is carried out with respect to various parameters used by different traditional works for H-type DWT. Fifty papers are considered for analysis. This analysis helps to find the significant parameters among all the parameters. The obtained results are graphically presented in Fig. 4. Power coefficients, TSR, wind speed, torque coefficient, drag force, dynamic stall, blade profile, moment of inertia, Reynolds number, turbulence intensity, pitch angle, blade number, vorticity, azimuthal angle etc. have been considered.

From Figure 9 it is found that power coefficients, TSR, wind speed, blade thickness, drag force and torque coefficient have been highly considered by existing studies at rate of 68%. Following this, rotor torque, generator and rotor speed, dynamic stall have been surveyed at a rate of 8%. Further, pitch angle, moment of inertia, blade number, blade profile, turbulence intensity, vorticity, azimuthal angle and Reynolds number have been regarded at a rate of 6%. Analysis revealed that only few studies considered vibration, dimple effect, noise generation, Chord to diameter ratio and blade aspect ratio at a rate of 2%. These analytical outcomes will serve as a guideline for assessing the H-DWT performance effectively for considering turbine designs accordingly.

SWOT analysis of the H-rotor is presented below and précised in Fig. 10.

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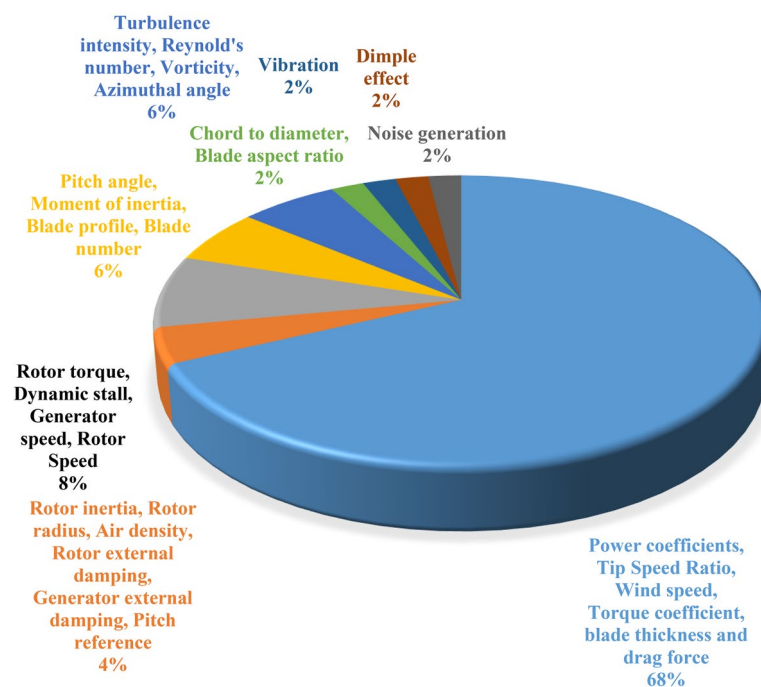


Fig. 9 Analysis of H-DWT with respect to various parameters

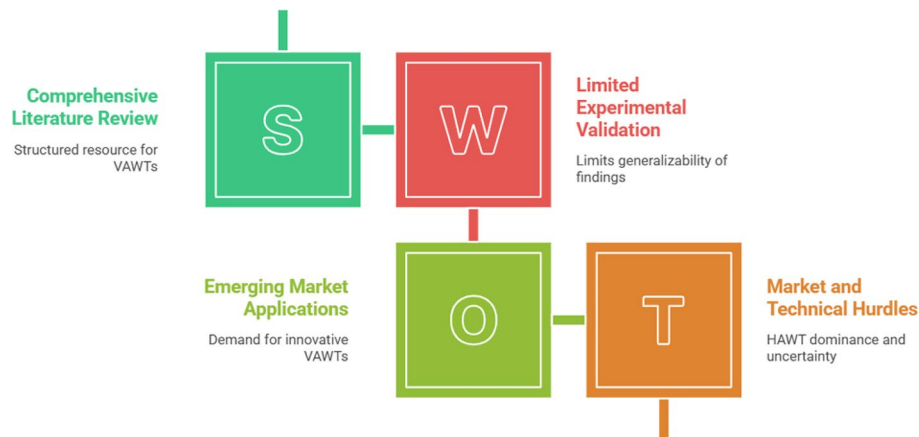


Fig. 10 SWOT analysis of the H-rotor

Strengths (S)

- Consolidates scattered literature on H-rotor aerodynamics, blade profiles, and CFD into a single, structured resource.
- Methodological depth: compares RANS vs. LES, turbulence models, grid strategies, and validation practices.
- Cross-links airfoil choice (e.g., NACA, NREL S-series) with performance metrics (C_p , TSR, dynamic stall behavior).
- Highlights urban/offshore suitability and omnidirectional operation unique to VAWTs.
- Identifies repeatable benchmarks (canonical cases, test rigs, tip-loss & end-effects treatments).

Weaknesses (W)

- Potential experimental scarcity versus CFD-heavy studies → limits generalizability.
- Inconsistent reporting across studies (Re, TSR, blockage, turbulence intensity) complicates meta-analysis.
- Limited coverage of fatigue/structural dynamics and control strategies relative to aerodynamics.
- Few standardized metrics for uncertainty quantification and grid independence in VAWT CFD.
- Possible underrepresentation of cost, manufacturability, and scalability (TRL gaps).

Opportunities (O)

- Growth of urban micro-wind and hybrid PV–VAWT markets; demand for low-noise, low-start-speed rotors.
- Advances in high-fidelity CFD (DES/LES), FSI, and GPU computing enable better dynamic stall capture.
- Additive manufacturing for rapid blade/profile iteration, including morphing/twist concepts.
- Open datasets/benchmarks & reproducible workflows (CFD cases, CAD, scripts).
- Policy and offshore VAWT pilots seeking compact, omnidirectional solutions.

Threats (T)

- Dominance of HAWTs in utility-scale projects; funding and benchmarks skewed toward HAWT tech.
- Noise, durability, and fatigue concerns in turbulent urban flows.
- Siting/standards gaps for VAWTs (certification, safety, power curve measurement).
- Variability in wind resource at micro-sites → performance uncertainty; skepticism from investors.
- Model risk: overreliance on 2D or steady-state models leading to optimistic predictions.

Applications and trends

VAWT have been utilised in rural and urban areas for commercial, industrial and residential applications. Installation of WTs in cities could assist in achieving a smooth energy demand. Recently, it is utilised for feeding commercial grid. In accordance with the resources of urban wind, the power production distribution has turned to be the significant issue. It could satisfy the requests for consumption in rural areas thereby subsidise to smart grid. VAWT and HAWT has been accepted in recent times. Huge number of WTs have been installed globally in last few decades. For instance: India and China have developed half of overall turbine installation throughout the world. All these corresponds to production in large-scale. As stated by WWEA (World Wind Energy Association) record, installation of WTs increased by 5% in comparison to early years. The estimated capacity of this turbine was higher than 945 MW that enhanced 14% in comparison to previous year. As VAWTs have been trustworthy for regionalised energy production, various system integrators and manufacturers have found significance in investment in this field. VAWT components and suppliers include urban-green energy, Hi-VAWT, Global-wing group, VBINE energy, clean-field energy etc. [91]. Similarly, installing H-type VAWT has limited cases by intrinsic start-up challenges related with this kind of turbine which could be vital in environments having minimum wind speed. H-type DWT has turned to be trending. Accordingly, CFD modelling has been afforded for analysing the start-up characteristics related with this kind of turbines. ANSYS Fluent has been employed for analysis. In contradiction to traditional work, turbine has been left free for acceleration in accordance with experience of torque over time. Comprehensive analysis has been accomplished considering grid generation and turbulence modelling that have been key to conform accuracy for this analysis [92]. Outcomes from simulation revealed better agreement based on empirical data. Techniques provide high accuracy level confirming its valuability for identical issues. Similarly, the CFD setup has been employed for assessing the impacts of various geometric features of turbine rotor on the initiating features. Cambered and symmetric airfoils having varied thickness with different pitch angles have been analysed. Ideal start-up features have been perceived with usage of NACA2418 showing 1.5 as pitch angle. This configuration minimised the start-up period while maintaining the peak performance of the turbine [57].

Research Gaps

Various gaps identified through analysis of the existing works are listed below.

- One of the major limitations of H-type Darrieus turbines is poor self-starting capability. Although several passive and active enhancement methods (e.g., variable pitch, helical blades, starter motors) have been proposed, comprehensive parametric studies quantifying their effects under different wind conditions are still insufficient.
- Experimental data, especially for dynamic stall, wake development, and self-starting behaviour of H-rotors are limited. Most field or wind tunnel tests focus on averaged quantities, neglecting transient behaviours. Furthermore, comparative validation of CFD simulations against time-resolved experimental results (e.g., via Particle Image Velocimetry) is lacking.
- Although several blade profiles (e.g., NACA 0018, NACA 0021, S-series) have been used, a systematic comparison of airfoils under dynamic stall and low TSRs is still lacking. Additionally, the influence of profile geometry on hysteresis behaviour and lift-drag characteristics in deep stall regimes requires further analysis.
- Most aerodynamic studies do not consider the impact of flow-induced vibrations and aeroelastic coupling in the rotor arms and blades. These factors can significantly affect fatigue life and operational stability, particularly at higher wind speeds or in off-design conditions.
- Research on hybrid H-rotor systems, such as those combining Savonius-Darrieus or dual-rotor H-type configurations, is limited. These systems show promise for self-starting and improved low-speed performance but lack detailed aerodynamic and structural investigations.
- While many CFD and performance evaluations are performed at optimal or rated wind speeds, the behaviour of H-type turbines under urban wind profiles, including turbulent and sheared inflows, is not well understood. Research into low Reynolds number aerodynamics, noise sensitivity, and turbine siting in built environments is limited.
- There is a lack of systematic studies employing multi-objective optimization techniques to design optimal blade shapes, aspect ratios, solidity, and strut configurations for minimizing drag while maximizing lift. Trade-offs between efficiency, noise emission, and structural integrity have rarely been investigated in a unified optimization framework.
- A majority of simulations still rely on two-dimensional CFD models, which are insufficient to capture complex flow features like blade tip vortices, arm-induced disturbances, and vertical wake behaviour in H-type Darrieus turbines. Full 3D CFD and Fluid-Structure Interaction (FSI) simulations are essential to account for the influence of structural flexibility and unsteady aerodynamics on performance.
- The application of bio-inspired blade geometries, flexible or morphing structures, and variable camber blades in H-type Darrieus turbines is an underexplored area. These concepts could enhance performance across a wider range of wind conditions and reduce fatigue loads.
- Most simulation models omit key resistive torque components such as generator load, shaft damping, and bearing friction, leading to discrepancies in angular velocity predictions. The lack of empirical data hampers the quantification of torque behavior during the startup phase. Furthermore, most studies utilize theoretical angle-of-attack values, whereas accurate modeling requires the effective angle-of-attack, especially under dynamic stall conditions.

- Noise generation mechanisms, wake–vortex–acoustic coupling, and mitigation strategies are largely unexplored, despite their importance for urban applications. Moreover, most validations are based on wind tunnel data; real-world field measurements under natural inflow conditions are scarce, leading to uncertainty in CFD-to-reality performance gaps.

Conclusion

The research aimed to analyse the aerodynamic performance of H-type DWT. Various factors influenced the efficacy of this turbine which was explored through this analysis namely ITV, blade thickness, pitch angle etc. Various designs have been considered by the traditional studies. Different simulations have been undertaken by them. This study tabulated the important design considerations and CFD simulations of conventional works encompassing the blade profiles such as NACA 0021, NACA4415, DU06W200, NACA0020, NACA0018, NACA0022, NACA634-221, NACA0015 with 2D and 3D CFD simulations. A critical analysis was undertaken by considering different parameters such as power coefficients, wind speed, blade thickness, drag force TSR, torque coefficient, pitch angle, blade number, blade profile, moment of inertia, turbulence intensity, vorticity, azimuthal angle etc. Analysis revealed that TSR, wind speed, drag force, blade thickness and torque coefficient was found to be highly considered at a rate of 68%. Whereas, vibration, dimple effect, Chord to diameter ratio, noise generation and blade aspect ratio was least considered. With respect to application, H-type DWT has various residential, economical and commercial benefits. However, it has various demerits which are also presented in this study. To solve these issues, this study suggested to undertake 3D simulations and consider significant parameters. It has also afforded other suggestions which can be considered by researchers for effective employment of this turbine in various applications.

Abbreviations

VAWT	Vertical Axis Wind Turbine
CMAES	Covariance Matrix Adaptation Evolutionary Strategy
TSR	Tip Speed Ratio
CFD	Computational Fluid Dynamics
BEM	Blade Element Momentum
WT	Wind Turbine
SST	Shear Stress Transport
GF	Gurney Flaps
HAWT	Horizontal Axis Wind Turbines
RANS	Reynolds-Averaged Navier–Stokes
LES	Large Eddy Simulation
BL	Baseline
FTC	Fault Tolerance Control
FDI	Fault Detection and Isolation
LPV	Linear Parameter Varying
LLCT	Linear Like Control Theory
NCD	Network Control Design
FAR	Finite Aspect Ratio
FW-H	Ffowcs Williams and Hawkings
MST	Megawatt Scale Turbines
VFM	Viscous Flow Model
ITV	Instantaneous Tangential Velocity
FSI	Fluid-Structure Interaction

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Authors' contributions

Venaganti Abhinaya: Writing Original draft, Methodology, Review and Editing. G Ganesh Kumar: Review and Editing, Supervision, Formal Analysis.

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Data availability

No Data available.

Declarations**Competing interests**

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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